Performance Evaluation of Keypoint-Based Models for Isolated Sign Language Recognition across Signer-Dependent, Signer-Independent, and Noisy Scenarios

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***Abstract*—Isolated Sign Language Recognition (ISLR) requires models that remain accurate across different signers, degraded inputs, and practical runtime constraints. This study compares four keypoint-based ISLR architectures: LSTM, CNN1D, GCN, and Transformer. All models use the same ASL Citizen hand-keypoint representation, preprocessing pipeline, and evaluation metrics. They are evaluated under signer-dependent (SD), signer-independent (SI), noisy-data, and computational-cost scenarios using 10-, 20-, and 30-frame inputs. Among the compared models, the Transformer achieves the highest observed clean-test accuracy at 30 frames (55.2% SD; 51.5% SI). Although CNN1D and GCN are faster, their recognition quality is lower, while LSTM has higher runtime cost. In noisy-data testing, the 30-frame Transformer achieves the highest Accuracy and Macro F1-score in both SD and SI scenarios. These findings reflect relative performance under the evaluated conditions, without establishing statistical consistency across training runs.**

***Index Terms*—Isolated Sign Language Recognition, Keypoint-Based Recognition, Model Comparison, Signer-Independent Evaluation, Robustness Evaluation.**

I. Introduction

Isolated Sign Language Recognition (ISLR) maps a pre-segmented video clip of one sign into a discrete vocabulary label. Compared with Continuous Sign Language Recognition (CSLR), ISLR provides a more controlled setting for architecture comparison because it avoids sign-boundary detection, temporal alignment, and sequence-decoding challenges [1]–[3]. This makes ISLR suitable for examining how different models recognize word-level sign gestures before moving toward more complex continuous-sign settings.

Sign language recognition remains difficult because signs depend on coordinated handshape, orientation, movement, and location, while signer variation can make visually similar signs harder to distinguish [4]–[7]. Keypoint-based ISLR is a practical alternative to raw RGB recognition because landmark-coordinate sequences reduce background dependence, lower input dimensionality, and support lightweight processing [8]–[10]. However, a keypoint representation alone does not determine which architecture is most reliable under deployment-oriented conditions.

Architectural choice is important because model families encode sign motion differently. LSTM models temporal dependencies recurrently, CNN captures local temporal patterns, GCN exploits hand-landmark topology, and Transformer uses self-attention to capture long-range temporal relations [11]–[13]. These inductive biases may behave differently when signer identity, input quality, or runtime constraints change. Therefore, a fair comparison should keep the dataset, preprocessing pipeline, input representation, and metrics fixed while varying only the model architecture.

This study uses ASL Citizen because it offers a large vocabulary, diverse contributors, and signer-aware partitioning that supports both signer-dependent (SD) and signer-independent (SI) evaluation [7]. SD evaluation can overestimate performance because signer identities may overlap between training and testing, whereas SI evaluation is stricter and closer to deployment because test signers are unseen [7], [9], [14]. Robustness and computation are also essential: low light, occlusion, motion blur, or unstable hand detection can degrade landmarks, and excessive inference time can limit practical use [15], [16].

This multi-scenario view follows the software-engineering principle that system quality should be assessed according to context of use. Multi-perspective evaluation has been applied

to academic website quality [17], while user-story conflict research shows that complex requirements need structured analysis to identify ambiguity and inconsistency before development decisions are made [18]. In keypoint-based ISLR, the same reasoning motivates evaluating clean recognition, cross-signer generalization, noisy-input robustness, and computational cost together rather than relying on a single accuracy score.

Although prior SLR studies have examined benchmarks, pose-based recognition, MediaPipe-based comparisons, skeleton models, graph models, Transformers, and video–pose fusion [1], [6], [9], [11]–[13], [16], [19], their findings leave room for controlled comparison of SD performance, SI generalization, noisy-data robustness, and computational cost within a single hand-keypoint pipeline. To address this gap, this study compares LSTM, CNN, GCN, and Transformer models using the same ASL Citizen hand-keypoint representation, preprocessing pipeline, and evaluation metrics.

The main contributions of this study are as follows:

- A controlled comparison of LSTM, CNN, GCN, and Transformer models using the same keypoint-based ISLR pipeline, dataset, and metrics.
- An empirical evaluation of SD and SI performance together with noisy-data robustness under low-light and occlusion conditions.
- An evaluation of trade-offs in recognition quality, cross-signer generalization, robustness, and computational cost.

This paper is organized as follows. Section II reviews related work, Section III describes the methodology, Section IV presents the results and discussion, and Section V concludes the paper.

II. Related Work

A. Isolated Sign Language Recognition

Sign languages convey meaning through coordinated movements of the hands and body, perceived through the visual system [5]. Recognition research divides into Continuous Sign Language Recognition (CSLR) and Isolated Sign Language Recognition (ISLR): whereas CSLR requires simultaneous segmentation and recognition of sign sequences, ISLR treats each video as a single-sign classification problem [2], [20], [21].

This simplicity makes ISLR a clean setting for comparing recognition architectures, since performance differences can be attributed to the model rather than to auxiliary sequence-level modules. Existing benchmarks span vocabulary sizes from small laboratory datasets with fewer than 100 classes to community-sourced corpora exceeding 2,000 classes [7], [14], and surveys consistently identify signer variability, dataset scale, and robustness under realistic conditions as the field's primary open challenges [22], [23].

B. Skeleton-Based Representation and Feature Extraction

Skeleton-based SLR represents signing as sequences of three-dimensional joint coordinates rather than raw pixels, which brings three practical advantages. The representation is compact: a 2-second clip needing about 6.44 MB as video fits in under 500 KB as a landmark sequence without losing the motion information needed for recognition [9]. It is invariant to background, lighting, and clothing, reducing exposure to spurious visual correlates. And it preserves signer privacy, since visual identity such as skin tone and face texture is not stored [10].

Recent ISLR work extracts these features with *MediaPipe Holistic* [8], which estimates body pose, facial landmarks, and hand keypoints from a single RGB frame: BlazePose provides the global pose, which then guides a region-of-interest crop that refines hand detection, yielding 543 normalized (x, y, z) landmarks [9], [10]. This study uses only the 21 landmarks per hand, which suffice to encode handshape, orientation, and motion trajectory—the parameters most directly tied to lexical sign identity [6], [7]—and keep the cross-architecture comparison focused on a single, well-defined representation. Fig. 2 shows these 21 hand landmarks.

C. Benchmark Datasets for Isolated Sign Language Recognition

The evaluation dataset largely determines whether benchmark results transfer to deployment with diverse, non-professional signers, and a signer-independent (SI) partition tests generalization far more rigorously than a signer-dependent (SD) one. Yet prior ISLR work often relied on datasets that are limited in scale or lack enforced SI splits: KArSL-100 and LSA64 have small signer pools that constrain SI generalization, while MS-ASL lacks reliable signer identity metadata. This motivates a benchmark offering both broad signer diversity and a construction-enforced SI protocol.

This study therefore uses *ASL Citizen* [7], a community-sourced benchmark of 83,399 isolated sign instances across 2,731 classes, recorded from 52 demographically diverse signers under varied conditions. It enforces a strict user-based split in which 11 test signers (about 21% of all signers) are never seen during training, so the evaluation measures genuine cross-signer generalization rather than memorization. The split can also be relaxed to the instance level for a direct SD-versus-SI comparison, and the dataset's scale supports evaluating multiple architectures (LSTM, CNN, GCN, and Transformer) without the overfitting risk of smaller benchmarks. For reference, Fig. 2 (left, labeled "original") shows a frame from the ASL Citizen dataset.

D. Signer-Dependent and Signer-Independent Evaluation

A critical distinction in ISLR evaluation is whether the test partition contains signers who also appeared in training, defining two protocols that measure different properties of the model.

In *signer-dependent* (SD) evaluation, training and test sets share signer identities, so the model has seen the motion style, body proportions, and idiosyncrasies of every test signer during training. It can thus exploit signer-specific shortcuts that do not generalize to unseen individuals, yielding high accuracies—often exceeding 95% on controlled datasets—that overestimate practical utility [9], [14].

In *signer-independent* (SI) evaluation, signer identities are entirely disjoint: no test signer contributed any training sample, forcing the model to generalize across natural variation in body size, movement speed, and signing dialect. SI accuracies are substantially lower, with reported drops of 15 to 30 percentage points relative to SD on the same dataset and architecture [9], [24]. This gap is the primary measure of deployment readiness, since real-world systems must recognize signers they have never encountered.

E. Deep Learning Architectures for Skeleton-Based ISLR

Recognizing isolated signs from skeleton data requires capturing both the spatial structure of the hands at each instant and its temporal evolution over the sign. We evaluate four distinct families of deep-learning architectures, each offering a different approach to learning from spatiotemporal landmark sequences.

1) LSTM: LSTM networks model temporal dependencies through recurrent memory mechanisms and have been widely used for skeleton-based sign recognition [9], [21]. However, spatial relationships among hand joints are learned implicitly because landmark coordinates are processed as feature vectors.

2) CNN1D: Temporal convolutional networks learn local motion patterns using one-dimensional convolutions applied along the temporal axis [11]. Their parallel computation and computational efficiency make them attractive for large-scale skeleton-based recognition tasks.

3) GCN: Graph Convolutional Networks explicitly represent skeletal structures as graphs, where landmarks are modeled as nodes and anatomical connections as edges [11], [25]. This representation enables direct learning of spatial relationships between hand joints.

4) Transformer: Transformer architectures employ self-attention mechanisms to capture global temporal dependencies and long-range relationships within sign sequences [12], [13], [24]. Recent studies have demonstrated strong performance on large-scale sign language benchmarks.

F. Robustness under Degraded Input Conditions

Although recent ISLR models achieve strong performance under controlled conditions, real-world deployment often involves imperfect visual input. Occlusion and low-light environments can reduce landmark quality, increase missing detections, and introduce spatial noise into skeleton representations [15], [19]. Previous studies have shown that such degradations can significantly affect recognition accuracy, motivating robustness evaluation as an important complement to standard benchmark performance [9], [24]. Accordingly, this study evaluates model robustness under occlusion and low-light conditions in addition to conventional SD and SI recognition settings.

III. Methodology

This research adopts a scenario-based methodology to evaluate isolated sign language recognition using hand-keypoint data. As shown in Fig. 1, the pipeline begins with RGB videos from the ASL Citizen dataset, extracts hand landmarks, converts them into standardized temporal tensors, trains the recognition models, and evaluates them under baseline, signer-generalization, and noisy-input scenarios.

A. Part I: Data Preparation and Scenario Setup

Experiments are conducted on the ASL Citizen dataset using the curated experimental split defined in this study for the signer-dependent/signer-independent and robustness scenarios. This working split contains isolated sign videos produced by multiple signers with diverse signing styles and includes a predefined set of sign classes with multiple samples per class [7]. Following the SD and SI protocols described in Section II-E, the data are partitioned into training, validation, and test subsets to compare performance on familiar signers and unseen signers. In the SI protocol, the training split contains 50,458 samples from 41 signers, while the test split contains 32,941 samples from 11 disjoint signers. In the SD protocol, the training split contains 50,456 samples and the test split contains 32,943 samples, with both partitions drawing from the same pool of 52 signers. Across all four files, the number of gloss classes remains fixed at 2,731, ensuring that SD and SI comparisons isolate signer overlap rather than label-space differences.

B. Part II: MediaPipe-based Feature Engineering and Preprocessing

Hand features are extracted from each video frame and represented as three-dimensional coordinates (x, y, z). For each frame t, the left- and right-hand landmarks are flattened and concatenated into a single vector $\mathbf{x}_t \in \mathbb{R}^D$, where D is fixed by the hand-only representation used throughout the study. Each video sample is then represented as the temporal matrix defined in Eq. 1:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T] \in \mathbb{R}^{T \times D}. \tag{1}$$

Fig. 2 shows an example visualization from the *ASL Citizen* dataset. The overlaid landmarks illustrate the hand-keypoint stream that is retained and transformed into the model input tensor in this study.

The preprocessing pipeline converts raw sign videos into model-ready tensors through several steps: landmark extraction, uniform sampling for sequence-length normalization to a fixed T, hand-landmark selection, coordinate normalization, missing-landmark handling, and padding with masking. This process ensures that all samples share a consistent temporal and feature structure, allowing all compared models to use the same standardized input representation.

C. Part III: Model Comparative

This study compares four model families for isolated sign language recognition: Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Graph Convolutional Networks (GCN), and Transformers, as discussed in Section II-E. To ensure a fair comparison, all models use the same standardized spatio-temporal keypoint tensors generated in the

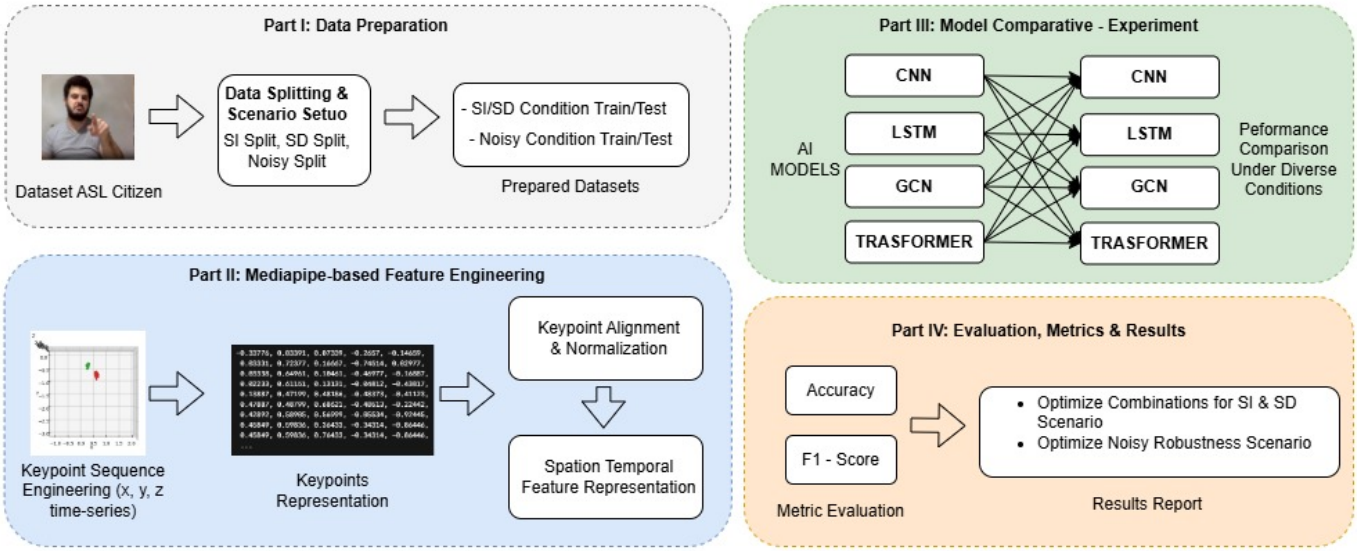


Fig. 1: Overall Research Workflow.

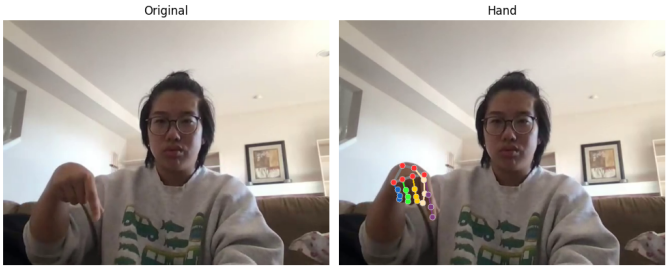


Fig. 2: Example of hand-keypoint extraction for the compared ISLR models.

previous phase, so performance differences can be attributed mainly to model design rather than data-format differences.

Model size is also compared through trainable parameter counts. Transformer is the largest model with 1,039,787 parameters, followed by LSTM with 880,229 parameters. CNN1D is more compact with 498,859 parameters, while GCN is the lightest model with 381,427 parameters. These parameter counts provide context for interpreting accuracy, robustness, and computational efficiency in the experiments.

D. Part IV: Evaluation Protocol and Results Reporting

The evaluation protocol combines standard classification metrics with a three-scenario experimental design. Metrics quantify recognition quality for each run, while the scenario-based setup isolates the specific factors influencing model performance.

1) Evaluation Metrics: Given the large-scale vocabulary of the ISLR task, the evaluation is treated as a multi-class classification problem using a one-vs-rest approach. The fundamental outcomes for each target sign are derived from the confusion matrix components:

- **True Positive** (TP)**:** A target sign is correctly identified when present.
- **False Positive** (FP)**:** A target sign is incorrectly identified when absent.
- **False Negative** (FN)**:** A target sign is missed when present.
- **True Negative** (TN)**:** The absence of a target sign is correctly identified.

Performance is primarily assessed using Accuracy to measure the overall ratio of correct predictions, formulated in Eq. 2:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

To mitigate the effect of class imbalance inherent in large datasets, macro-averaging is applied to the F1-score, which provides the harmonic mean of Precision ($\frac{TP}{TP+FP}$) and Recall ($\frac{TP}{TP+FN}$). The Macro-F1 is defined by Eq. 3:

$$\text{Macro-F1} = \frac{1}{C}\sum_{c=1}^{C} 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{3}$$

where C is the total number of sign classes. All metrics are computed consistently across every model family and experimental scenario.

2) Scenario-Based Experimental Design: All models are evaluated under three controlled scenarios, summarized in Table I. The baseline comparison scenario evaluates the intrinsic modeling capacity of different architectures (LSTM, CNN, GCN, and Transformer) using identical hand-keypoint representations, preprocessing procedures, and training settings. The signer generalization scenario isolates signer-related effects by comparing Signer-Dependent (SD) and Signer-Independent (SI) data splits while keeping the model configurations unchanged. Finally, the robustness test examines model behavior under realistic input degradation, specifically low-light and occlusion conditions, using the same hand-keypoint extraction pipeline and evaluation metrics.

TABLE I: Scenario-based experimental design.

Scenario	Objective	Variable
Baseline Comparison	Compare performance across model families.	Model type
Signer Generalization	Assess performance on unseen signers.	SD vs SI split
Robustness Test	Evaluate resilience under degraded conditions.	Low-light, occlusion

Robustness is evaluated on 13,012 degraded test videos, leaving the training data unchanged. The low-light condition is produced by reducing brightness, contrast, and saturation while increasing the vignette effect. Occlusion is simulated using localized blur covering portions of the signing region. Fig. 3 shows representative examples of the noisy input data used in this test scenario.



Fig. 3: Representative noisy test samples generated using low-light adjustment and occlusion.

IV. Results and Discussion

All experiments use Categorical Crossentropy with SGD and are evaluated using Accuracy and Macro F1-score. Sections 4.1–4.3 report the 10-frame, 100-epoch baseline with learning rate 0.01, batch size 32, seed 42, and the stated CPU/GPU setup. Section 4.4 analyzes 20- and 30-frame inputs, followed by noisy-data robustness testing.

A. Training Process on 10-Frame Samples

The 10-frame training process is summarized briefly to keep the results section focused on nonredundant evidence. Across SD and SI settings, the Transformer shows the strongest convergence pattern, while GCN and CNN1D improve more slowly and remain in a lower-performance region. LSTM learns more effectively than CNN1D and GCN, but its final behavior remains below the Transformer. Thus, the 10-frame, 100-epoch setting provides a stable baseline for the testing results and longer-frame analysis.

B. Testing Evaluation on 10-Frame Samples

Table II summarizes the test performance of the four model families under the 10-frame hand-only configuration for both signer-dependent (SD) and signer-independent (SI) scenarios. The table reports test accuracy, test F1-score, and test loss to provide a compact comparison of predictive performance and optimization behavior across evaluation settings.

Table II shows that the Transformer is the strongest model in both SD and SI settings. The bolded values identify the best result for each metric because the Transformer achieves

TABLE II: Test results under SD and SI scenarios on 10 frame samples across 100 training epochs

Scenario	Metric	CNN1D	GCN	LSTM	Transformer
SD	Test Acc	0.156634	0.147983	0.248581	**0.411377**
	Test F1	0.138659	0.134865	0.231201	**0.405521**
	Test Loss	4.911060	4.918382	3.891554	**2.972796**
SI	Test Acc	0.158708	0.139704	0.238639	**0.386965**
	Test F1	0.141911	0.125559	0.224887	**0.382210**
	Test Loss	5.056445	5.141787	4.015034	**3.105853**

Note: Bold values mark the best model for each scenario–metric row; higher Accuracy/F1 is better, while lower Loss is better.

the highest Accuracy and Macro F1-score, where larger values indicate better recognition quality, and the lowest test loss, where a smaller value indicates a more favorable prediction objective. LSTM is second, while CNN1D and GCN remain substantially lower, indicating that local convolution or hand-landmark graph structure alone does not match the sequence-level representation learned by self-attention.

The Transformer's advantage also persists when testing shifts from familiar signers to unseen signers. This is important because SI evaluation is closer to deployment, where signer style, body proportion, and motion rhythm differ from training users [7], [14]. Its higher Accuracy, higher Macro F1-score, and lower loss indicate stronger overall correctness, better class-level balance, and a more stable predictive distribution. This pattern is consistent with prior skeleton-based SLR studies showing that attention-based or Transformer-based models can focus on discriminative temporal regions and long-range landmark dependencies [11]–[13].

Overall, SI remains the more deployment-relevant setting because it exposes cross-signer generalization rather than signer overlap. The Transformer is the only model that maintains relatively strong performance in both SD and SI, supporting its use as the strongest 10-frame baseline before robustness and efficiency are considered.

C. Computational Complexity Analysis

In addition to predictive performance, model efficiency is also important for practical deployment. The methodology section summarizes the number of trainable parameters for the four model families under the same 10-frame hand-only configuration: Transformer has 1,039,787 parameters, LSTM has 880,299, CNN1D has 498,859, and GCN has 381,427. These values show that the best-performing model is also the largest one in terms of parameter count, while GCN and CNN1D are lighter alternatives with lower representational capacity.

Table III summarizes the 10-frame inference-time comparison by model and scenario. To provide a complete baseline efficiency view, the table reports mean, minimum, and maximum inference time in milliseconds for both SD and SI settings.

Table III shows that runtime is not determined by parameter count alone. CNN1D is the fastest model at 4.502/5.385 ms

TABLE III: Inference-time comparison across models under the 10-frame baseline.

Model	Scenario	Mean Time (ms)	Min Time (ms)	Max Time (ms)
CNN1D	SD	**4.502**	**4.260**	**4.901**
	SI	**5.385**	**4.340**	6.343
GCN	SD	6.030	5.261	7.025
	SI	5.410	5.056	**5.659**
LSTM	SD	28.105	25.584	35.204
	SI	23.813	22.473	24.693
Transformer	SD	12.101	11.242	12.636
	SI	12.527	11.227	14.579

for SD/SI, followed closely by GCN at 6.030/5.410 ms. These results show that lightweight convolutional and graph-based pipelines process short hand-only sequences very efficiently.

When recognition quality is prioritized, however, the Transformer offers the best 10-frame trade-off. Although it has the largest parameter count, its mean inference time remains moderate at 12.101/12.527 ms for SD/SI, far below LSTM's 28.105/23.813 ms. Thus, the Transformer provides the strongest Accuracy and Macro F1-score without an excessive latency penalty, while LSTM is weaker in both accuracy and runtime.

The SD–SI timing differences are small for CNN1D, GCN, and Transformer, suggesting that the signer split mainly affects recognition quality rather than inference cost. LSTM shows the widest absolute runtime range because recurrent computation is slower overall.

D. Effect of Increasing Frame Samples

This subsection extends the discussion beyond the 10-frame baseline by examining how longer temporal inputs affect model behavior when the frame length is increased to 20 and 30 frames.

1) Frame-Length Comparison: The first comparison across frame lengths discusses how the transition from 10 frames to 20 and 30 frames changes the training and testing results of the four model families. To preserve the information previously shown in the graph while avoiding visual duplication, Table IV reports the main Accuracy and Macro F1-score values in tabular form.

Table IV shows that training performance generally increases as the temporal window becomes longer, but the corresponding testing gains depend strongly on architecture. CNN1D and GCN obtain only small test improvements despite higher training scores, indicating limited generalization benefit from longer hand-keypoint sequences.

LSTM and Transformer benefit more clearly from additional context. LSTM improves to 0.408/0.364 test Accuracy and 0.395/0.355 Macro F1-score at 30 frames for SD/SI, while the Transformer remains best at every frame length, reaching 0.552/0.515 Accuracy and 0.546/0.511 Macro F1-score at 30 frames. This pattern shows that longer inputs increase representational power, but the strongest gains are concentrated in the attention-based architecture.

TABLE IV: Train and test performance across 10-, 20-, and 30-frame configurations.

Model	Fr.	SD Acc. Tr./Te.	SI Acc. Tr./Te.	SD F1 Tr./Te.	SI F1 Tr./Te.
CNN1D	10	0.265/0.157	0.285/0.159	0.250/0.139	0.274/0.142
CNN1D	20	0.338/0.184	0.350/0.183	0.327/0.169	0.343/0.170
CNN1D	30	0.355/0.186	0.370/0.182	0.345/0.171	0.363/0.168
GCN	10	0.216/0.148	0.222/0.140	0.203/0.135	0.210/0.126
GCN	20	0.261/0.172	0.274/0.177	0.249/0.160	0.265/0.166
GCN	30	0.254/0.166	0.266/0.162	0.243/0.155	0.256/0.152
LSTM	10	0.383/0.249	0.399/0.239	0.366/0.231	0.387/0.225
LSTM	20	0.571/0.364	0.593/0.339	0.561/0.350	0.588/0.327
LSTM	30	0.629/0.408	0.648/0.364	0.622/0.395	0.644/0.355
Transformer	10	**0.753/0.411**	**0.761/0.387**	**0.752/0.406**	**0.761/0.382**
Transformer	20	**0.830/0.516**	**0.842/0.486**	**0.829/0.509**	**0.841/0.481**
Transformer	30	**0.850/0.552**	**0.862/0.515**	**0.849/0.546**	**0.862/0.511**

Note: Each metric cell is reported as Train/Test.



Fig. 4: Training instability of the Transformer model under the longer-frame SD configurations.

For context, the original ASL Citizen benchmark reports 59.52% SI top-1 accuracy for pose-based ST-GCN and 63.10% for RGB-based I3D, above our 51.5% SI result [7]. These results are not directly comparable because our models use hand-only keypoints with 10–30 sampled frames, whereas the published baselines employ different input representations and temporal processing.

2) Training Stability Analysis: Fig. 4 shows that the longer-frame SD Transformer runs are sensitive to random seed. One run enters a normal learning trajectory, whereas another remains nearly flat, indicating that the same configuration can either converge or stall depending on initialization.

This instability is better interpreted as inconsistent optimization rather than evidence that the Transformer cannot use longer sequences. The seed-128 runs converged more normally than the seed-42 runs at 20 and 30 frames, showing that the architecture has high potential but a higher training-risk profile.

Therefore, longer inputs should be judged not only by final Accuracy and F1-score, but also by convergence consistency across seeds. For the Transformer, additional frames raise the observed performance ceiling but also increase optimization sensitivity. Systematic repeated-seed comparisons were not performed across all models; thus, ranking consistency and

TABLE V: Mean inference time across frame lengths in SD/SI settings.

Model	10 Fr.	20 Fr.	30 Fr.	Main Observation
CNN1D	4.50/5.38	5.18/4.38	6.32/4.57	Low and stable
GCN	6.03/5.41	6.08/6.63	6.08/6.37	Low and stable
LSTM	28.10/23.81	37.94/37.12	56.27/51.08	Highest increase
Transformer	12.10/12.53	12.06/12.84	14.94/11.45	Moderate growth

Note: Values are SD/SI mean inference times in milliseconds.

statistical significance remain unverified.

3) Computational Cost Across Frame Lengths: The final frame-length comparison examines whether the additional temporal context from 20- and 30-frame inputs introduces a proportional runtime cost. Table V provides the computational counterpart to Table IV using mean single-video inference time.

Runtime scaling is architecture-dependent. CNN1D and GCN remain almost flat across frame lengths, while Transformer increases only modestly from about 12 ms at 10–20 frames to 14.94/11.45 ms at 30 frames for SD/SI.

LSTM scales least efficiently, rising from 28.10/23.81 ms at 10 frames to 56.27/51.08 ms at 30 frames for SD/SI. Combined with Table IV, this supports the conclusion that Transformer gains the most from longer inputs while keeping a far better runtime profile than LSTM.

E. Noisy-Data Robustness Evaluation

This subsection evaluates the trained models under the low-light and occlusion procedures described in Section III. Crucially, degradation is applied only to the test videos, while the training data and the learned weights remain unchanged. This test-time-only design evaluates recognition under degradation without noise-augmented retraining. The low-light condition lowers brightness, contrast, and saturation, whereas the occlusion condition introduces localized blur and gray rectangular masks over part of the signing region; both reduce the reliability of MediaPipe hand detection and inject spatial noise into the keypoint stream. Table VI summarizes Accuracy, Macro F1-score, and mean prediction confidence (Conf.) across models, scenarios, and frame lengths.

Across the 13,012-video degraded-input evaluation, the reported results describe recognition performance under the tested perturbations. Across both SD and SI scenarios and all three frame lengths, the Transformer obtains the highest Accuracy and Macro F1-score. Such perturbations can affect hand-pose estimation, as reported in prior work [15].

The 30-frame Transformer records the strongest noisy-data results in both scenarios, reaching 0.3423 Accuracy and 0.3340 Macro F1-score under SD, and 0.3321 Accuracy and 0.3234 Macro F1-score under SI. For this model, both metrics improve as the input window increases from 10 to 20 and then 30 frames. Thus, the 30-frame Transformer has the highest observed noisy-data Accuracy and Macro F1-score under these conditions.

TABLE VI: Noisy-data testing results across SD and SI scenarios.

Sc.	Model	Fr.	Acc.	F1	Conf.
Signer-Dependent	CNN1D	10	0.0895	0.0768	0.2187
	GCN	10	0.0804	0.0713	0.1708
	LSTM	10	0.1413	0.1293	0.2326
	Transformer	10	**0.2378**	**0.2328**	**0.4331**
	CNN1D	20	0.1075	0.0960	0.2744
	GCN	20	0.0984	0.0884	0.1987
	LSTM	20	0.2044	0.1925	0.3688
	Transformer	20	**0.3169**	**0.3067**	**0.5396**
	CNN1D	30	0.1093	0.1010	0.2939
	GCN	30	0.0946	0.0832	0.1922
	LSTM	30	0.2350	0.2215	0.4098
	Transformer	30	**0.3423**	**0.3340**	**0.5404**
Signer-Independent	CNN1D	10	0.1014	0.0872	0.2354
	GCN	10	0.0826	0.0723	0.1711
	LSTM	10	0.1456	0.1325	0.2464
	Transformer	10	**0.2418**	**0.2340**	**0.4119**
	CNN1D	20	0.1161	0.1040	0.2827
	GCN	20	0.1101	0.0981	0.2104
	LSTM	20	0.2097	0.1959	0.3717
	Transformer	20	**0.3185**	**0.3085**	**0.5189**
	CNN1D	30	0.1603	0.1522	0.4373
	GCN	30	0.1769	0.1652	0.3932
	LSTM	30	0.1970	0.1880	0.5287
	Transformer	30	**0.3321**	**0.3234**	**0.5461**

Note: The degraded-input evaluation uses 13,012 test videos. Bold indicates the highest value within each frame group and scenario. Higher Acc./F1 is better. Conf. denotes mean prediction confidence.

Among the other architectures, LSTM generally achieves the next-highest Accuracy and Macro F1-score, while CNN1D and GCN trail the Transformer; the relative ranking of CNN1D and GCN varies by scenario and frame length. The Transformer also has the highest mean prediction confidence in each SD/SI frame group, but confidence alone does not establish prediction calibration or reliability. These results support the value of effective temporal modeling when skeleton observations are incomplete [19]. Overall, the 30-frame Transformer is the strongest configuration in this noisy-data evaluation.

V. Conclusion

This study evaluated LSTM, CNN1D, GCN, and Transformer models for keypoint-based ISLR under signer-dependent, signer-independent, noisy-data, and computational-cost scenarios. The Transformer achieves the highest observed clean-test Accuracy and Macro F1-score among the compared models at 30 frames (55.2% SD; 51.5% SI). The SI results confirm that cross-signer generalization is a critical deployment-oriented evaluation dimension, while the stability analysis shows that longer-frame Transformer training can become sensitive to random initialization.

From a computational and robustness perspective, CNN1D and GCN are the fastest models but have substantially lower recognition quality, while LSTM benefits from longer inputs at the cost of the highest runtime. The Transformer combines moderate inference time with the highest observed noisy-data Accuracy and Macro F1-score at 30 frames in both SD and SI.

As systematic multiple-seed comparisons were not conducted, consistency of the reported rankings remains unverified. Future work should address training stability and evaluate broader degradation conditions.

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